$$\begin{cases}\mathbf{e}_{1|A} = a_{11}\,\mathbf{e}_{1|B} + a_{12}\,\mathbf{e}_{2|B} + a_{13}\,\mathbf{e}_{3|B}, \\ \mathbf{e}_{2|A} = a_{21}\,\mathbf{e}_{1|B} + a_{22}\,\mathbf{e}_{2|B} + a_{23}\,\mathbf{e}_{3|B}, \\ \mathbf{e}_{3|A} = a_{31}\,\mathbf{e}_{1|B} + a_{32}\,\mathbf{e}_{2|B} + a_{33}\,\mathbf{e}_{3|B}.\end{cases}$$

(1)

$e_1 e_2 e_{3A} = a_{11} a_{12} a_{13} a_{21} a_{22} a_{23} a_{31} a_{32} a_{33AB} e_1 e_2 e_{3B}$

(2)

$e_{1|A}, e_{2|A}, e_{3|A}$

A

$e_{1|B}, e_{2|B}, e_{3|B}$

B

$e_{i|A}$

B

$e_{j|B}$

$a_{ij} \mathbf{e}_{i|A} \cdot \mathbf{e}_{j|B} (i, j = 1, 2, 3)$

(3)

$A a_{11} a_{12} a_{13} a_{21} a_{22} a_{23} a_{31} a_{32} a_{33}$

(4)

$\mathbf{v}_{|A} \mathbf{v}_{|B} AB \mathbf{v} AB$

$\mathbf{v}_{|A} = A\mathbf{v}_{|B}$

$e_1 e_2 e_{3A} = a_{11} a_{12} a_{13} a_{21} a_{22} a_{23} a_{31} a_{32} a_{33AB} e_1 e_2 e_{3B}$

$\{\ |e_{1|A}| = \sqrt{a_{11}^2 + a_{12}^2 + a_{13}^2} = 1, |e_{2|A}| = \sqrt{a_{21}^2 + a_{22}^2 + a_{23}^2} = 1, |e_{3|A}| = \sqrt{a_{31}^2 + a_{32}^2 + a_{33}^2} = 1.$

$\{\ e_{1|A} \cdot e_{2|A} = a_{11} a_{21} + a_{12} a_{22} + a_{13} a_{23} = 0, e_{1|A} \cdot e_{3|A} = a_{11} a_{31} + a_{12} a_{32} + a_{13} a_{33} = 0, e_{2|A} \cdot e_{3|A} = a_{21} a_{31} + a_{22} a_{32} + a_{23} a_{33} = 0.$

$a_{11} a_{12} a_{13} a_{21} a_{22} a_{23} a_{31} a_{32} a_{33AB} = a_{22} a_{23} a_{32} a_{33} - a_{21} a_{23} a_{31} a_{33} a_{21} a_{22} a_{31} a_{32} - a_{12} a_{13} a_{32} a_{33} a_{11} a_{13} a_{31} a_{33} - a_{11} a_{12} a_{31} a_{32} a_{12} a_{1}$

(5)

$e_1 e_2 e_{3A} = a_{11} a_{12} a_{13} a_{21} a_{22} a_{23} a_{31} a_{32} a_{33AB} e_1 e_2 e_{3B}$

$e_{1|A} = e_{2|A} \times e_{3|A} = e_{1|B} e_{2|B} e_{3|B} a_{11} a_{12} a_{13} a_{21} a_{22} a_{23} = a_{22} a_{23} a_{32} a_{33} e_{1|B} - a_{21} a_{23} a_{31} a_{33} e_{2|B} + a_{21} a_{22} a_{31} a_{32} e_{3|B}$

(6)

$A^{-1} = A^T$

$A^T A = I$

$\mathbf{v}_{|A} \mathbf{v}_{|B} AB \mathbf{v} AB$

$\mathbf{v}_{|A} = A\mathbf{v}_{|B}$

$\mathbf{v}_{|A} \mathbf{v}_{|B}$

$\mathbf{v}_{|A} \cdot \mathbf{v}_{|A} = \mathbf{v}_{|B} \cdot \mathbf{v}_{|B}$

$\mathbf{v}_{|B}^T A^T A \mathbf{v}_{|B} = \mathbf{v}_{|B}^T \mathbf{v}_{|B}$

$\mathbf{v}_{|B}$

$A^T A = I$

$A^{-1} = A^T$

$\mathbf{v}$

$\mathbf{v}_{|A} \tilde{v}_{|A}$

$\mathbf{v}_{|B} \tilde{v}_{|B}$

$\mathbf{v}_{|B} \tilde{v}_{|B}$

$\tilde{v}_{|A} = A_{AB} \tilde{v}_{|B} A_{AB}^{\mathrm{T}}$

$A_{AB}^{\mathrm{T}} \tilde{v}_{|A} = \tilde{v}_{|B} A_{AB}^{\mathrm{T}}$

(7)

$\tilde{v}_{|A} A_{AB} = A_{AB} \tilde{v}_{|B}$

(8)

A_{ij}

$\lambda \lambda =$

$A_{ij} \lambda$

$f_A(\lambda) \det(A - \lambda I_3)$

$= \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix}$

$a_{22}+$
$a_{33} =$
$\mathrm{tr}(A)$

$$= -\lambda^3 + \mathrm{tr}(A)\,\lambda^2 - \left(\sum_{k=1}^{3} M_{kk}\right)\lambda + \det(A).$$

$A^T A =$
I
$M_{11}+M_{22}+M_{33} = \mathrm{tr}(A),$

$\det A =$
1

$$f_A(\lambda) = -\lambda^3 + \mathrm{tr}(A)\,\lambda^2 - \mathrm{tr}(A)\,\lambda + 1.$$

$\lambda =$
1

$$f_A(1) = -1 + \mathrm{tr}(A) - \mathrm{tr}(A) + 1 = 0.$$

$\lambda =$
$1 f_A(\lambda) A$

$$A\lambda = \lambda \tag{9}$$

λ

$\theta\lambda$

$$a_{11} = \cos\theta + \lambda_1^2(1-\cos\theta) a_{12} = -\lambda_3 \sin\theta + \lambda_1\lambda_2(1-\cos\theta) a_{13} = \lambda_2 \sin\theta + \lambda_3\lambda_1(1-\cos\theta) a_{21} = \lambda_3 \sin\theta + \lambda_1\lambda_2(1-\cos\theta) a_{22} = \mathrm{c} \tag{10}$$

$\delta_{ij}, \epsilon_{ijk}$

$$\delta_{ij} 1 - \frac{1}{4}(i-j)^2[5-(i-j)^2] (i,j = 1,2,3) \tag{11}$$

$$\epsilon_{ijk} \frac{1}{2}(i-j)(j-k)(k-i) (i,j,k = 1,2,3) \tag{12}$$

$$a_{ij} = \delta_{ij}\cos\theta - \epsilon_{ijk}\lambda_k \sin\theta + \lambda_i\lambda_j(1-\cos\theta) (i,j = 1,2,3) \tag{13}$$

$e_{i|A} =$
$A e_{i|B} a_{ij} =$
$e_{i|A} \cdot$
$e_{j|B} =$
$e_{i|B} \cdot$
$(A e_{j|B})$
$_rotation_formula$

$$e_A = e_B \cos\theta - (e_B \times \lambda)\sin\theta + \lambda(\lambda \cdot e_B)(1-\cos\theta)$$

$\lambda_i \lambda \cdot$
$e_{i|B} =$
$\lambda \cdot$
$e_{i|A}$
??$\alpha_1, \alpha_2, \alpha_3 I I_{ij|A}$

$$I_{ij|A} \alpha_i \cdot I \cdot \alpha_j (i,j = 1,2,3)$$

$I_{ij|A} (i,j =$
$1,2,3)$
$b_1, b_2, b_3 \alpha_1, \alpha_2, \alpha_3$
$I_{ij|B} b_i \cdot$
$I \cdot$
$b_j (i,j =$
$1,2,3)$
$b_1 b_2 b_3$

$$I_{12|B} = I_{21|B} = I_{23|B} = I_{32|B} = I_{31|B} = I_{13|B} = 0$$

$$I_{11|B} = \frac{m}{12}(12^2+3^2)L^2 = \frac{153}{12} m L^2$$

$$I_{22|B} = \frac{m}{12}(3^2+4^2)L^2 = \frac{25}{12} m L^2$$

$$I_{33|B} = \frac{m}{12}(4^2+12^2)L^2 = \frac{160}{12} m L^2$$

$$A_{ij} = I_3 \cos\theta + \tilde{n} \sin\theta + n n^{\mathrm{T}} (1 - \cos\theta). \tag{14}$$

$$\omega_{ij|j} = n\dot{\theta} + \sin\theta\, \dot{\tilde{n}} - (1 - \cos\theta)\tilde{n}\dot{n}. \tag{15}$$

n^{T}

$$\dot{\theta} = n^{\mathrm{T}} \omega_{ij|j}. \tag{16}$$

$$\dot{n} = \frac{1}{2\sin\theta} \left[\sin\theta\, I_3 - (1 + \cos\theta)\tilde{n}\right] \tilde{n}\omega_{ij|j} = \frac{1}{2} \left[I_3 - \cot(\theta/2)\, \tilde{n}\right] \tilde{n}\omega_{ij|j}.$$

λ
θ
n
θ
$LABBABA$
$aAbBa$

$$b = a\cos\theta - (a \times \lambda)\sin\theta + \lambda(\lambda \cdot a)(1 - \cos\theta) \tag{17}$$

$\tilde{\lambda} a =$
$\lambda \times a$

$$b = \left[\cos\theta\, I + \sin\theta\, \tilde{\lambda} + (1 - \cos\theta)\, \lambda\lambda^{\mathrm{T}}\right] a. \tag{18}$$

$$AU \cos\theta - (U \times \lambda)\sin\theta + \lambda(\lambda \cdot U)(1 - \cos\theta) \tag{19}$$

$$b = a \cdot A \tag{20}$$

$\alpha_1 \alpha_2 A$

$\alpha_1 \parallel L_A$

$\alpha_2 = \lambda \times \alpha_1$

$\beta_1 \beta_2 B$

$\beta_1 \parallel L_B$

$\beta_2 = \lambda \times \beta_1$

$\theta = 0$

$a = b$

$\alpha_1 = \beta_1$

$\alpha_2 = \beta_2$

ab

$a = p\alpha_1 + q\alpha_2 + r\lambda$

$b = p\beta_1 + q\beta_2 + r\lambda$

$\beta_1 = \cos\theta \alpha_1 + \sin\theta \alpha_2$

$\beta_2 = -\sin\theta \alpha_1 + \cos\theta \alpha_2$

$_rotation_formula$

$$A_{ij} = I_3 \cos\theta + \tilde{n} \sin\theta + n n^{\mathrm{T}} (1 - \cos\theta). \tag{21}$$

$$\cos\theta = 2\cos^2\frac{\theta}{2} - 1, \sin\theta = 2\cos\frac{\theta}{2}\sin\frac{\theta}{2}, (1 - \cos\theta) = 2\sin^2\frac{\theta}{2}. \tag{22}$$

$$[\varepsilon_1 \varepsilon_2 \varepsilon_3]^{\mathrm{T}}\, n \sin\frac{\theta}{2} =: \tilde{\varepsilon}, \varepsilon_4 \cos\frac{\theta}{2}, \tag{23}$$

$$A_{ij} = I_3(2\varepsilon_4^2 - 1) + 2\tilde{\varepsilon}\varepsilon_4 + 2\tilde{\varepsilon}\tilde{\varepsilon}^{\mathrm{T}}. \tag{24}$$

??

$$\rho_i = \frac{\epsilon_i}{\epsilon_4}(i=1,2,3)$$

A

$$A = \frac{1}{1+\rho_1^2+\rho_2^2+\rho_3^2} 1+\rho_1^2-\rho_2^2-\rho_3^2 2(\rho_1\rho_2-\rho_3) 2(\rho_3\rho_1+\rho_2) 2(\rho_1\rho_2+\rho_3) 1+\rho_2^2-\rho_3^2-\rho_1^2 2(\rho_2\rho_3-\rho_1) 2(\rho_3\rho_1-\rho_2) 2(\rho_2\rho_3+\rho_1) 1+\rho_3^2$$

??a
b

$\mathbf{a}-\mathbf{b} = (\mathbf{a}+\mathbf{b})\times\rho$

(88)

$$\rho = \frac{(\alpha_1-\beta_1)\times(\alpha_2-\beta_2)}{(\alpha_1+\beta_1)\cdot(\alpha_2-\beta_2)}$$

α_i
$\beta_i i =$
$1,2$
$\alpha_i =$
$\beta_i i =$
$1,2$

$$\rho_1 = \frac{a_{32}-a_{23}}{1+a_{11}+a_{22}+a_{33}}, \rho_2 = \frac{a_{13}-a_{31}}{1+a_{11}+a_{22}+a_{33}}, \rho_3 = \frac{a_{21}-a_{12}}{1+a_{11}+a_{22}+a_{33}}$$

$A_{ij} = I_3\cos\theta+\tilde{n}\sin\theta+nn^{\mathrm{T}}(1-\cos\theta).$

(89)

$\tilde{n}$
i
j

$\dot{A}_{ij} = \left(-I_3\sin\theta+\tilde{n}\cos\theta+nn^{\mathrm{T}}\sin\theta\right)\dot{\theta}.$

(90)

$\tilde{\omega}_{ij|j} =$
$A_{ij}^{\mathrm{T}}\dot{A}_{ij}$
$=$
$\left[I_3\cos\theta+\tilde{n}^{\mathrm{T}}\sin\theta+nn^{\mathrm{T}}(1-\cos\theta)\right]\left(-I_3\sin\theta+\right.$
$\tilde{n}\cos\theta+$
$\left.nn^{\mathrm{T}}\sin\theta\right)\dot{\theta}$

$\tilde{n}\tilde{n} =$
$nn^{\mathrm{T}}-$
I_3

$\tilde{\omega}_{ij|j} =$
$A_{ij}^{\mathrm{T}}\dot{A}_{ij}$
$=$
$\left[I_3\cos\theta+\tilde{n}^{\mathrm{T}}\sin\theta+nn^{\mathrm{T}}(1-\cos\theta)\right]\left(-I_3\sin\theta+\right.$
$\tilde{n}\cos\theta+$
$\left.nn^{\mathrm{T}}\sin\theta\right)\dot{\theta}$
$=$
$\left(-I_3\sin\theta\cos\theta+\right.$
$\tilde{n}\cos^2\theta+$
$\left.nn^{\mathrm{T}}\sin\theta\cos\theta\right)\dot{\theta}$
$+$
$\left(-\tilde{n}^{\mathrm{T}}\sin^2\theta+\tilde{n}^{\mathrm{T}}\tilde{n}\sin\theta\cos\theta+\tilde{n}^{\mathrm{T}}nn^{\mathrm{T}}\sin^2\theta\right)\dot{\theta}$
$+$
$\left[-nn^{\mathrm{T}}\sin\theta(1-\cos\theta)+nn^{\mathrm{T}}\tilde{n}\cos\theta(1-\cos\theta)+nn^{\mathrm{T}}nn^{\mathrm{T}}\sin\theta(1-\cos\theta)\right]\dot{\theta}$
$=$
$\tilde{n}(\cos^2\theta+$
$\sin^2\theta)\dot{\theta}$
$+$
$\left[-I_3+nn^{\mathrm{T}}-(nn^{\mathrm{T}}-I_3)\right]\sin\theta\cos\theta\dot{\theta}$
$=$
$\tilde{n}\dot{\theta}$
$=$
$\widetilde{n\dot{\theta}}.$

$\omega_{ij|j} = n\dot{\theta}.$

(91)

$\tilde{n}$
i
j

$\omega_{ij|j} = n\dot{\theta}.$

(92)

$\omega_{ij}^i =$
$A_{ij}\omega_{ij|j}$
$=$
$\left[I_3\cos\theta+\tilde{n}\sin\theta+nn^{\mathrm{T}}(1-\cos\theta)\right]n\dot{\theta}$
$=$
$n\dot{\theta} =$
$\omega_{ij|j}.$

$\omega_{ij}^i = \omega_{ij|j} = n_{ij}\dot{\theta}_{ij}.$